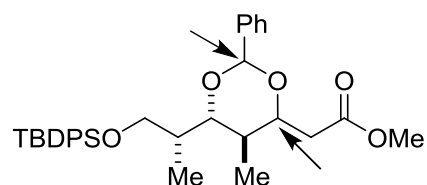
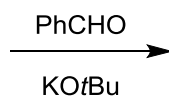
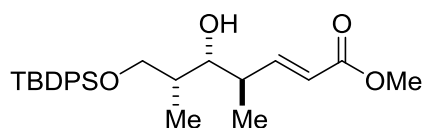
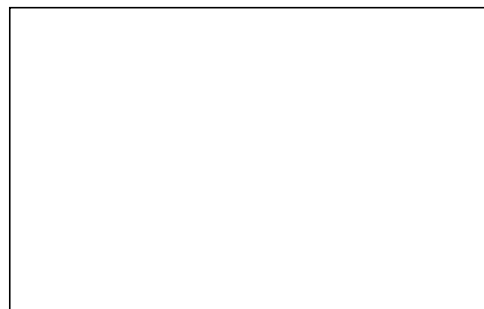
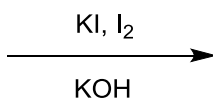
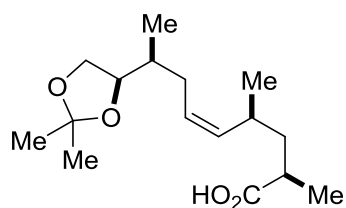
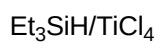
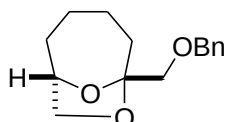
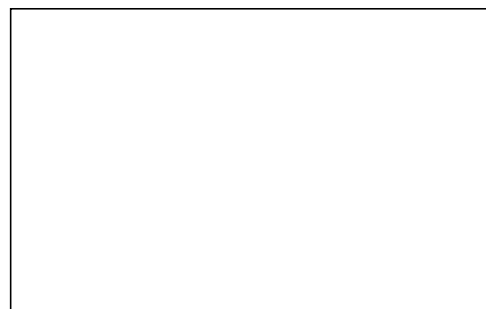
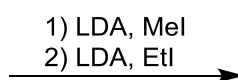
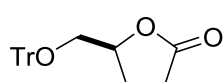


Exercises

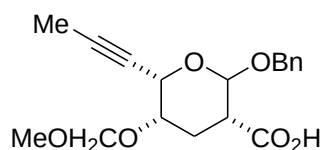
1. Predict the stereochemical outcome of the following transformations:



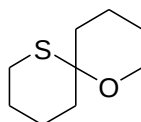
Stereochemistry?



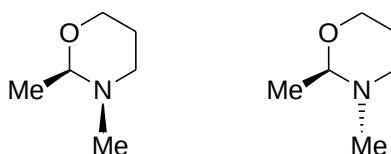
2. Draw the preferred conformation. Show all stabilizing interactions.



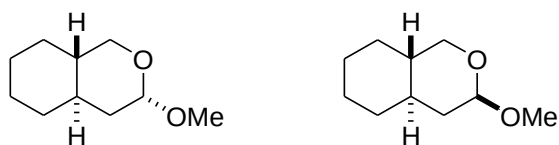
3. Draw all 4 possible chair conformers of the following spiro compound. Sort them by their relative stability and justify your order by appropriate arguments and drawings.



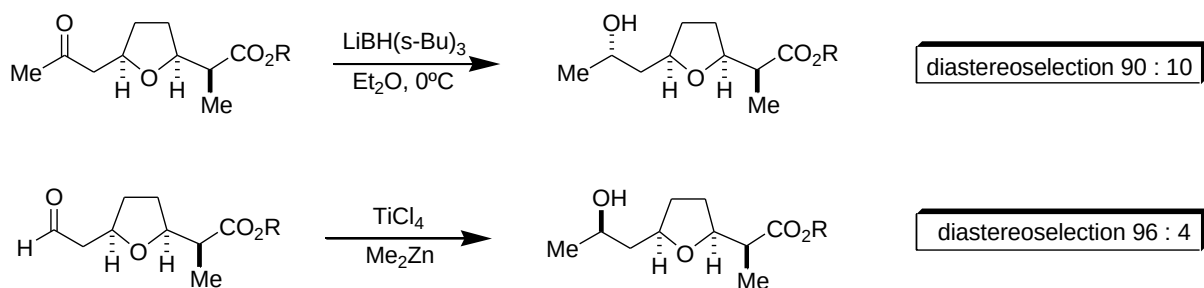
4. By low-temperature NMR studies it was found that one of the two conformers shown is more stable. Which one is it and why? Why was the system studied at low temperature?



5. The two acetals shown are hydrolyzed with different reaction rate. What is the reaction product? Which isomer is hydrolyzed faster and why?

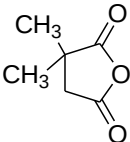


6. Explain the following experimental results:



7. Give a mechanistic rationale for the observed stereochemical outcome:

Reduction of Representative Cyclic Unsymmetrical Anyhrides with LiHBR_3 in Tetrahydrofuran at 0°C

Anhydride	Reagent	Products, ratio %	
	LiAlH_4	5	95
	NaBH_4	-	sole product
	LiEt_3BH	25	75